# ADM  LT 2025/2026: Hackathon

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## 1 Problem Context and Objective

In this work, a query-answering system is developed for a hackathon task based on structured textual evidence. The objective is to answer natural language questions using the facts provided for each database. The notebook loads the training, development, and test data, extracts the evidence corpus, prepares the retrieval and generation components, trains a query-type router, defines specialized answer functions, evaluates the pipeline on the development set, and creates the final submission file. The system is designed to handle the different query types required by the task: Lookup, Boolean, Count, Aggregation, and Set operations.

## 2 Methodological Framework

The approach is based on a query-type-aware pipeline. First, each query is classified using a router trained with TF-IDF features and Logistic Regression. The predicted query type determines which answer function is applied.

For evidence selection, the notebook implements a hybrid retrieval strategy. It combines BM25 lexical retrieval with dense retrieval using BGE embeddings. The two rankings are merged through reciprocal rank fusion, and the selected candidates are reranked using a MiniLM cross-encoder. This allows the system to focus on the most relevant facts before generating or extracting the final answer.

For answer generation and extraction, the notebook combines model-based and rule-based strategies. A Qwen2.5-Instruct model with LoRA adaptation is used for Lookup questions, Count fallback cases, Aggregation fallback cases, and generic Set operation cases. The notebook also documents the LoRA fine-tuning procedure, where training examples are constructed from Lookup, Count, and Set operation queries using BM25-selected facts and query-type-specific prompts. FLAN-T5 is used for Boolean questions and as an additional fallback component for Aggregation.

## 3 Conceptual Contributions and Key Design Principles

The main idea is that the task should not be solved with a single generic answering method. Each query type has a different structure, so the notebook uses specialized logic depending on the predicted category.

A second key idea is that retrieval quality is essential. Since the answer must be derived from the available facts, the system first reduces the evidence space through lexical search, semantic retrieval, fusion, and reranking.

Finally, the notebook combines language models with deterministic rules. This is important because some queries require flexible text understanding, while others require stricter operations such as counting entities, comparing numeric values, resolving aggregation questions, or returning a set of answers in a controlled format.